

Step 1: Import Libraries

```
In [2]: import numpy as np
```

Step 2: Load Dataset

```
In [4]: data = np.genfromtxt(
        "SS.csv",
        delimiter=",",
        skip_header=1
    )

print(data[:5])

[[101.  89.  76.  91.  95.]
 [102.  65.  72.  70.  88.]
 [103.  91.  94.  87.  98.]
 [104.  55.  61.  58.  80.]
 [105.  82.  78.  85.  92.]]
```

Step 3: Separate Columns

```
In [5]: student_id = data[:,0]

        maths = data[:,1]

        science = data[:,2]

        english = data[:,3]

        attendance = data[:,4]
```

Step 4: Basic Statistics

Total Marks

```
In [20]: print(np.sum(maths))
        print(np.sum(science))
        print(np.sum(english))

382.0
381.0
391.0
```

Average Marks

```
In [6]: print(np.mean(maths))
        print(np.mean(science))
        print(np.mean(english))

76.4
76.2
78.2
```

Highest Marks

```
In [7]: print(np.max(maths))

91.0
```

Lowest Marks

```
In [8]: print(np.min(science))

61.0
```

Median

```
In [9]: print(np.median(english))

85.0
```

Standard Deviation

```
In [10]: print(np.std(maths))
```

14.079772725438433

Variance

```
In [11]: print(np.var(science))
```

113.75999999999999

Step 5: Total Marks

```
In [22]: total = 382 + 381 + 391
print(total)
```

1154

Step 6: Percentage

```
In [24]: per = total / 3
print(per)
```

384.6666666666667

Step 7: Grade Calculation

```
In [31]: grade = np.where(percentage>=90, "A+",
                        np.where(percentage>=80, "A",
                        np.where(percentage>=70, "B",
                        np.where(percentage>=60, "C", "Fail"))))

print(grade)
```

A+

Step 8: Top Performers

```
In [32]: top_students = student_id[percentage>=90]
print(top_students)
```

[[101. 102. 103. 104. 105.]]

Step 9: Failed Students

```
In [37]: failed = student_id[percentage<40]
print(failed)
```

[]

Step 10: Attendance Analysis

Average Attendance

```
In [38]: np.mean(attendance)
```

```
Out[38]: np.float64(90.6)
```

Students Below 75%

```
In [39]: student_id[attendance<75]
```

```
Out[39]: array([], dtype=float64)
```

Step 11: Subject Comparison

```
In [40]: highest_subject = np.argmax([
    np.mean(maths),
    np.mean(science),
    np.mean(english)
])

print(highest_subject)
```

Step 12: Rank Students

```
In [41]: rank = np.argsort(-percentage)
print(rank)
```

```
[0]
```

```
In [ ]:
```